

DIT5307 Brief Development Report: THEi HR System

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1. Introduction

The **THEi HR System** is a Java EE 8 web application for HR administration — managing employees, training enrollments, instalment-based fee payments, and automated payment-overdue warnings. Built with **EJB 3.2**, **JPA 2.2** and **JSF 2.3**; packaged as a Maven WAR and deployed on **GlassFish 5.1**.

2. Enterprise Session Beans

CrudServiceBean — **@Stateless**. Centralises persistence for Employee, TrainingEnrollment, FeePaymentRecord and PaymentWarning (create / find / list / update / deactivate-warning, plus JPQL queries such as findEmployeeByHkid and getActiveWarnings). *Stateless* because every CRUD call is conversation-free, allowing the container to pool a small number of instances for many concurrent JSF requests with no stale per-client state.

TimerServiceBean — **@Singleton @Startup**. Owns the EJB Timer Service. `createPaymentWarningTimer(enrollment)` schedules a single-action timer for 30 calendar days before `trainingStartDate`; the `@Timeout` callback then checks `fullyPaid`, sets `paymentOverdue=true`, and persists a `PaymentWarning`. *Singleton* guarantees exactly one bean handles every timer expiry — avoiding duplicate warnings and race conditions; `@Startup` bootstraps it with the application.

3. Transaction Management

All EJB methods run under **Container-Managed Transactions** with the default **REQUIRED** attribute, so each public method on `CrudServiceBean` and the `@Timeout` callback execute atomically. Concrete boundaries: (a) `EnrollmentDetailController.recordPayment()` calls `addPaymentRecord()` + `updateTrainingEnrollment()` together so the new instalment row and the enrollment's running total commit as one unit; (b) `deactivateWarningForEnrollment()` is a JPQL bulk `UPDATE` that atomically flips all warnings for one enrollment to `active=false`; (c) `TimerServiceBean.timeout()` sets `paymentOverdue` and persists the `PaymentWarning` in the same transaction, keeping the flag and warning row consistent.

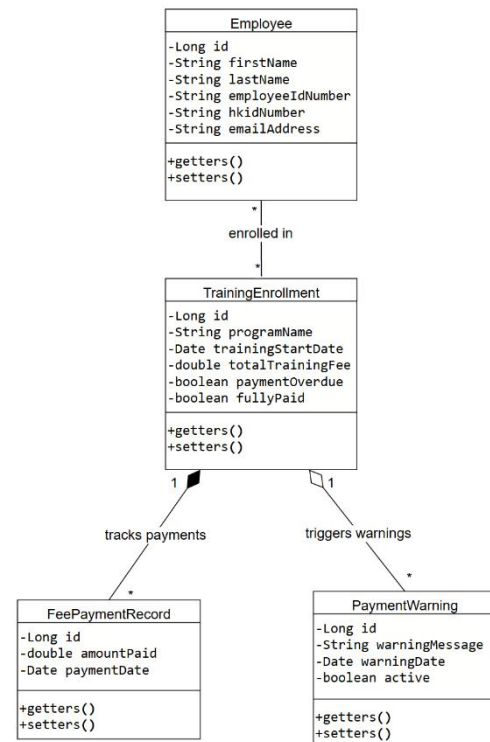
4. JSF Pages and Backing Beans

- **login.xhtml / LoginController** (`@SessionScoped`) — authenticates the hard-coded HR account.
- **index.xhtml / DashboardController** — retrieves active `PaymentWarnings` via `getActiveWarnings()` and renders them **sorted ascending by days remaining to the 21-day cancellation deadline**; overdue entries display “Payment Overdue” only.
- **employees.xhtml / EmployeeController** — lists / creates Employees (title MR/MS/MRS/DR, dept, mobile, email, HKID).
- **enrollment.xhtml / EnrollmentController** — lists and creates enrollments; on create, invokes `TimerServiceBean.createPaymentWarningTimer()`.
- **editEnrollment.xhtml / EditEnrollmentController** — edits program info, training dates and enrolled employees.
- **enrollmentDetail.xhtml / EnrollmentDetailController** — records instalment payments, shows payment history, and provides the “**Mark Fully Paid**” action that clears `paymentOverdue` and deactivates related warnings.

5. System Design (UML)

Relationships. Employee ↔ TrainingEnrollment is **Many-to-Many** (join table ENROLLMENT_EMPLOYEE). TrainingEnrollment → FeePaymentRecord is **One-to-Many** (cascade=ALL, orphanRemoval=true). TrainingEnrollment ↔ PaymentWarning is **One-to-One**.

Class summary (key fields only): - **Employee** — id, lastName @NotNull, firstName @NotNull, title (MR/MS/MRS/DR), employeeIdNumber {unique}, department, mobilePhoneNumber @Pattern(^[\0-9]{8}\$), emailAddress @Email, hkidNumber {unique}. - **TrainingEnrollment** — id, programName, programDescription, enrollmentDate, trainingStartDate, totalTrainingFee, paymentOverdue, fullyPaid. - **FeePaymentRecord** — id, amountPaid, paymentDate, @ManyToOne trainingEnrollment. - **PaymentWarning** — id, warningMessage, warningDate, active, @OneToOne associatedEnrollment.



6. JPA Highlights

Bean Validation is enforced via @NotNull, @Email, @Pattern (mobile = 8 digits), @Size, plus DB-level unique/nullable=false on employeeIdNumber and hkidNumber. **Key JPQL queries** in CrudServiceBean: SELECT e FROM Employee e WHERE e.emailAddress = :email; SELECT e FROM Employee e WHERE e.hkidNumber = :hkid; SELECT w FROM PaymentWarning w WHERE w.active = true (with cache REFRESH hint so newly-fired warnings appear immediately); and a bulk UPDATE PaymentWarning w SET w.active=false WHERE w.associatedEnrollment.id = :eid.

7. Assumptions

A single hard-coded HR admin account is sufficient (per the brief). For the demo, createPaymentWarningTimer() is shortened to **30 seconds** instead of 30 days before start date — the production line is retained in the source and commented next to the demo override. Days-remaining used for warning sort = trainingStartDate - today - 21 (matches the 21-day cancellation cut-off). All dates use the **Asia/Hong Kong** timezone; currency uses standard decimal (HKD). One PaymentWarning per enrollment is sufficient; re-marking unpaid reuses the existing row, so duplicates are not created.

8. References

1. Apache Software Foundation, “Apache Maven project documentation,” 2023. Available: <https://maven.apache.org/>
2. Oracle, “Java EE 8 Technologies (EJB 3.2, JPA 2.2, JSF 2.3, etc.).” Available: <https://www.oracle.com/java/technologies/java-ee-8.html>
3. Oracle, “Java EE 8 SDK Downloads (includes GlassFish 5.0.1).” Available: <https://www.oracle.com/java/technologies/javaee-8-sdk-downloads.html>
4. Oracle, “Java EE Tutorial — Using the EJB Timer Service.” Available: <https://docs.oracle.com/javaee/7/tutorial/ejb-basicexamples004.htm>